

SC 5401
WASSCE 2026
GENERAL
SCIENCE 1
Objective Test
1½ hours

1

Name

Index Number

THE WEST AFRICAN EXAMINATIONS COUNCIL

West African Senior School Certificate Examination
for School Candidates

SC 2026

GENERAL SCIENCE 1
OBJECTIVE TEST
[60 marks]

1½ hours

Do **not** open this booklet until you are told to do so. While you are waiting, read and observe the following instructions. Write your **name** and **index number** in the spaces provided above.

1. Use **2B** pencil throughout.
2. On the pre-printed answer sheet, check that the following details are **correctly** printed:
 - (a) In the space marked **Name**, check your **surname** followed by your **other names**.
 - (b) In the spaces marked *Examination*, *Year*, *Subject* and *Paper*, check 'WASSCE (SC)', '2026', 'GENERAL SCIENCE', and '1' respectively.
 - (c) In the box marked *Index Number*, your **index number** has been printed vertically in the spaces on the left-hand side, and **each** numbered space has been shaded in line with **each** digit. **Reshade** each of the shaded spaces.
 - (d) In the box marked *Subject Code*, the digits 540113 are printed vertically in the spaces on the left-hand side. **Reshade** the corresponding numbered spaces as you did for your index number.
3. An example is given below. This is for a male candidate whose name is Tony Mensah ATIAH. His *index number* is 7102143958 and he is offering *General Science 1*.

THE WEST AFRICAN EXAMINATIONS COUNCIL

ANSWER SHEET

PRINT IN BLOCK LETTERS	
Name: ATIAH TONY MENSAH	GHA
Examination: WASSCE (SC)	Year: 2026
Subject: GENERAL SCIENCE	Paper: 1

INSTRUCTIONS TO CANDIDATES	
1.	Use grade 2B pencil throughout.
2.	Answer each question by choosing one letter and shading like this: A ☒ B ☐ C ☐ D ☐
3.	Erase completely any answer you wish to change.
4.	Leave extra spaces blank if the answer spaces provided are more than you need.
5.	Do not make any markings across the heavy black marks at the right hand edge of your answer sheet.

INDEX NUMBER										SUBJECT CODE												
7	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	5	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	
1	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	4	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	
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3	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9	For Supervisors only If candidate is absent shade this space ☐											
9	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9												
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8	☐ 0	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	☐ 8	☐ 9												

Answer **all** the questions.

Each question is followed by **four** options lettered **A** to **D**. Find the **correct** option for **each** question and shade in **pencil** on your answer sheet, the answer space which bears the same letter as the option you have chosen.

Give only **one** answer to **each** question. An example is given below.

Which of the following elements is a metal?

- A. Carbon
- B. Copper
- C. Helium
- D. Krypton

The correct answer is **Copper**, which is lettered **B**, and therefore answer space **B** would be shaded.

☐ A ☒ B ☐ C ☐ D

Think carefully before you shade the answer spaces; erase completely any answers you wish to change.

Do all rough work in this question paper.

Now answer the following questions.

1. Which of the following processes is responsible for the passage of genetic material from parent to the offspring in plants?
 - A. Budding
 - B. Fertilization
 - C. Pathogenesis
 - D. Pollination

2. Benefits Ghana stands to gain in using solar panels instead of fossil fuels to generate electricity include
 - I. energy independence,
 - II. lowering of greenhouse effect and climate change,
 - III. improvement in air quality.
 - A. I only
 - B. II and III only
 - C. I and II only
 - D. I, II and III

3. When a force acts on a body, the body could
 - I. start moving.
 - II. stop moving.
 - III. change its direction.
 - A. I and II only
 - B. II and III only
 - C. I and III only
 - D. I, II and III

4. The long nature of the Loop of Henle helps
- A. to enhance the concentration of the urine.
 - B. re-absorption of glucose back into the blood.
 - C. re-absorption of amino acid back into the blood.
 - D. ultra-filtration taking place in the Bowman's capsule.
5. In the production of local soap, the burnt cocoa husk acts as
- A. alkali.
 - B. acid.
 - C. salt.
 - D. fragrance.
6. A doctor who checks for movement, sensation and reflexes in a patient is most **likely** suspecting damage to the
- A. brain ligament.
 - B. muscles.
 - C. tendon.
 - D. spinal cord.
7. Which of the following practices may lead to lifestyle disease?
- A. Eating unhealthy diet
 - B. Having insufficient physical activity
 - C. Consuming alcohol
 - D. Eating contaminated food
8. The **major** product formed during the manufacture of local soap could be classified as an organic
- A. alkali.
 - B. acid.
 - C. acid-salt.
 - D. salt.
9. Which of the following conditions helps the skin to excrete urea **faster**?
- A. Eating high protein
 - B. Exposing oneself to wet environment
 - C. Exposing oneself to high temperature
 - D. Drinking less water
10. Three resistors of $10\ \Omega$, $15\ \Omega$ and $20\ \Omega$ are connected in series. Calculate the effective resistance.
- A. $26.0\ \Omega$
 - B. $45.0\ \Omega$
 - C. $20.8\ \Omega$
 - D. $6.0\ \Omega$
11. The electronic component best suitable to convert alternating current from the mains to a radio which operates on direct current is
- A. transformer.
 - B. inductor.
 - C. capacitor.
 - D. p-n junction diode.

12. Which of the following materials has **negative** effect on soap produced locally?
- Presence of glycerol by-product
 - Excess alkali
 - Residual woodash
 - Colour of fat/oil used
- I, II and III only
 - II, III and IV only
 - I, III and IV only
 - I, II, III and IV
13. Which of the following impurities should be used in doping a tetravalent semi-conductor to increase the number of holes?
- Phosphorus
 - Boron
 - Silicon
 - Sulphur
14. Despite the repeated number of sex education programmes mounted by school authorities for their boarding students, the problem of drug abuse has persisted. Which of the following factors is **likely** to be responsible for the challenge faced by the authorities?
- Lack of family involvement in the control measure
 - The academic qualification of the students
 - The mental health of the students
 - The influence of peer pressure

Use the statement below to answer questions 15 and 16.

A stock solution of hydrochloric acid is labelled 10.0 M HCl.

15. What volume of the stock solution is required to prepare 250 cm^3 of 0.50 M HCl?
- 25.0 cm^3
 - 15.5 cm^3
 - 12.5 cm^3
 - 10.5 cm^3
16. Calculate the dilution factor for the prepared solution.
- 25
 - 20
 - 15
 - 10
17. The lung's ability to excrete CO_2 is because the diffusion of the gas
- produces toxic effect in the blood.
 - is from the alveoli into the blood.
 - is from the blood into the alveoli.
 - is produced during tissue respiration.
18. Which of the following blood groups would **not** make a man the father of a child of blood group O?
- Group A
 - Group B
 - Group AB
 - Group O

19. Which of the following household activities do **not** involve diffusion?
- Slow deflation of a balloon
 - Dissolving sugar in tea
 - Detecting the smell of food in the kitchen
 - Getting the hands wrinkled after putting it in hot water over a prolonged time
- I and IV only
 - II and III only
 - I, II and III only
 - I, II, III and IV
20. A p-n junction diode is connected in a circuit with the p-type connected to the positive terminal of a cell, and the n-type to the negative terminal, with a light emitting diode and a switch. The effect on the light emitting diode when the switch is closed is that the light emitting diode
- comes on because charges flow to it.
 - does not come on since the junction widens.
 - comes on since the junction narrows.
- I only
 - I and II only
 - I and III only
 - II and III only
21. Which of the following scenarios involves **both** the central nervous system and the peripheral nervous system? -
- Blinking when something gets close to the eye
 - Playing musical instruments at a function
 - Pupils constricting when exposed to bright light
 - Sneezing when an irritant enters the nose
22. Which of the following scientific principles is **not** associated with the production of indigenous beer and gin?
- Saponification
 - Fermentation
 - Distillation
 - Milling
23. An electric appliance rated 2000 W is used for 5 hours. Calculate the energy consumed within this time in kWh.
- 10
 - 400
 - 10,000
 - 100,000
24. Which of the following pairs of hereditary diseases in humans is associated with the blood?
- Albinism and sickle cell
 - Colour blindness and diabetes
 - Haemophilia and sickle cell
 - Colour blindness and haemophilia
25. Blood groups are **not** conclusive in determining the paternity of a child because
- blood groups of men change over a period of time.
 - the blood group of a child also changes with time.
 - many people share the ABO and Rh antigens.
 - the blood antigens in humans are also found in the rhesus monkey.

26. Which of the following Indigenous industries does **not** involve fermentation process?
- Soap making
 - Production of beverages
 - Production of gari
 - Production of kenkey
27. *The noble gases are used to create inert atmosphere.* This statement implies that the gases are
- unstable.
 - nonreactive.
 - rare.
 - reactive.
28. Which of the following solids could be classified as a non-metal electrical conductor?
- Aluminium
 - Copper
 - Carbon graphite
 - Carbon diamond
29. Which of the following substances could be used as a drying agent for carbon(IV) oxide?
- CaO
 - Dil. H_2SO_4
 - Conc. HCl
 - Conc. H_2SO_4
30. A man heterozygous for the albino gene marries a woman of the same genotype. The proportion of the children with the phenotype of the parents would be...
- 25%
 - 50%
 - 75%
 - 100%

Use the information below to answer questions 31 to 33.

Separate pieces of red and blue litmus papers were placed into solutions labelled X and Y. The table below shows the observations made.

Study the table and answer the questions that follow.

Litmus paper	Colour in X	Colour in Y
Red	Red	Red
Blue	Red	Blue

31. Solution X is likely to be identified as
- liquid in car battery.
 - solution of washing soda.
 - solution of table salt.
 - distilled water.
32. The pH of solution Y is likely to be ...
- 0
 - 3
 - 7
 - 10

33. Which of the following pairs of chemical substances are **likely** to have the same observations as X and Y respectively?
- Table salt and vinegar
 - Dilute hydrochloric acid and table salt
 - Vinegar and washing soda
 - Hydrogen peroxide and liquid in car battery
34. An out-patient reports of tingling muscle, weakness and numbness. Which of the following conditions should the doctor be investigating?
- Nerve compression
 - Diabetes
 - Peripheral neuropathy
- I only
 - I and II only
 - II and III only
 - I, II and III
35. Which of the following diseases is **likely** to have *shortness of breath, wheezing and cough* as symptoms?
- Asthma
 - Pneumonia
 - Tuberculosis
- I only
 - I and II only
 - II and III only
 - I, II and III
36. Which of the following statements is **true** about menstruation?
- Absence of menstruation always indicate active pregnancy
 - During menstruation only the layers of endometrium is shed
 - In a normal 28-day menstrual cycle, menstruation lasts 3-5 days
- I and II only
 - II and III only
 - I and III only
 - I, II and III
37. The malfunctioning of a patient's kidney has been traced to damage to the nephron which has resulted in producing urine which is darker in colour with more salts. Which part of the nephron may have suffered the **most** damage?
- The malpighian body
 - The distal convoluted tubule
 - The Loop of Henle
 - The collecting duct
38. A lady's inability to conceive after several years of marriage has been traced to the husband after repeated tests on the lady and the husband. Which of the following statements about the man may **not** be correct?
- The man produces semen without sperms
 - Low sperm count may account for the condition
 - The seminiferous tubules of the testes may have been blocked
 - The man produces large number of sperms than normal

39. Which of the following statements **best** describes the events leading up to the pollen tube entering the embryo sac in a flowering plant? The pollen tube grows
- through the style carrying the male gamete.
 - on the stigma and grows through the style carrying two male gametes.
 - towards the embryo sac carrying the two gametes and the endosperm nuclei.
 - through the ovary and the style carrying two male gametes.
40. Fishermen at sea ran out of drinking water and decided to drink sea water to quench their thirst. Which of the following statements **best** explains the effect of the sea water on the fishermen?
- Their body cells will absorb a lot of water from the sea water
 - Their body cells will lose water and dehydrate
 - The salt in the sea water will be removed by the kidney converting it to fresh water
 - There will be no effect on the fishermen
41. The use of steel in the construction industry depends on its
- strength.
 - durability.
 - lustre.
 - conductivity.
- I and II only
 - II and III only
 - III and IV only
 - I and IV only
42. The **correct** sequence of impulse transmission in a reflex arc is
- effector→sensory nerve→relay neuron→motor neuron
 - receptor→sensory neuron→relay neuron→motor neuron→effector
 - receptor→motor neuron→relay neuron→effector
 - sensory neuron→receptor→motor neuron→relay neuron→effector
43. Electric charge of 1800 coulombs move through a potential difference of 240 V. Determine the energy generated.
- 2.4×10^5 J
 - 4.3×10^5 J
 - 5.4×10^5 J
 - 3.3×10^5 J
44. An N-P-N transistor is used in household appliances **mainly** for
- amplification.
 - switching.
 - shutting down.
- I only
 - II and III only
 - I and II only
 - I, II and III
45. A convex lens of focal length 20 cm forms an image of an object at infinity. Determine the distance of the image from the lens.
- 10 cm
 - 20 cm
 - 30 cm
 - 40 cm

46. Which of the following factors affect the voltage in a secondary coil of a transformer?
- Number of turns in the secondary
 - The input voltage
 - Non-lamination of the core
- I only
 - II and III only
 - I and II only
 - I, II and III
47. Insanitary conditions in a community have been identified as the cause of ill-health of residents leading to various diseases and infections. Which of the following actions is most appropriate to improve the personal hygiene of residents?
- Ensuring access to clean drinking water
 - Construction of latrines and waste disposal systems
 - Promoting safe food preparation and storage
 - Taking collective action to become open-defecation free
48. Boxes containing different substances labelled with hazard symbols have been displayed in a community. Which of the following household chemicals could bear a hazard symbol?
- Ethanol
 - Insecticide
 - Toilet polish
 - Nail polish
49. A body of mass 5 kg is acted upon by a force, F and moves with a velocity of 10 m s^{-1} to a velocity of 30 m s^{-1} in 2 s. Calculate F .
- 10 N
 - 30 N
 - 50 N
 - 150 N
50. A rainbow is formed in the sky as a result of
- refraction of light by suspended rain drops.
 - dispersion of light by suspended rain drops.
 - reflection of light by suspended rain drops.
- I only
 - II only
 - I and II only
 - II and III only
51. Which of the following statements about a falling object is correct? A falling object has
- constant acceleration.
 - constant velocity.
 - constant displacement.
 - constant speed.

52. A body of density 25 kg m^{-3} has volume 10 m^3 . Determine the upthrust on it when it floats in salt water. [$g = 10 \text{ m s}^{-2}$, density of salt water = 1030 kg m^{-3}]
- 25.0 N
 - 250.0 N
 - 100.0 N
 - 2500.0 N
53. A bullet of mass 0.005 kg was shot from a pistol of mass 5.0 kg. If it moves out with a velocity of 100 m s^{-1} , determine the velocity and direction of the gun.
- 0.1 m s^{-1} backwards
 - 0.1 m s^{-1} forwards
 - 0.5 m s^{-1} backwards
 - 0.5 m s^{-1} forwards
54. Which of the following pairs of diseases are **exclusively** associated with the respiratory system?
- Cancer and Asthma
 - Pneumonia and Asthma
 - Asthma and Hepatitis
 - Tuberculosis and Cancer
55. The features which adapt the alveolus to its functions include the
- large surface area,
 - thin wall,
 - rich blood supply capillaries,
 - moist surface.
- I and II only
 - II and III only
 - III and IV only
 - I, II, III and IV
56. One **major** importance of indigenous industries is the
- improvement of local economies and provision of income opportunities.
 - reduction in population growth.
 - transfer of technology from different cultures.
 - movement of goods and materials across cultures.
57. The following observations are associated with fermentation process in gari making **except**
- preservation of material.
 - nutritional enhancement.
 - development of taste and aroma.
 - increasing size with water.
58. The following complications may affect a mother during pregnancy **except**
- high blood pressure while in labour.
 - gestational diabetes.
 - labour that begins before 37 weeks of pregnancy.
 - pelvic inflammatory disease.

59. A fish is able to see 180° above water. When rays of light from the fish's eye hits the water surface at a certain angle θ° , the refracted rays lie on the interface. The angle θ is referred to as the
- A. emergent angle.
 - B. critical angle.
 - C. angle of dispersion.
 - D. angle of reflection.
60. An athlete runs a 400 m race in 40 s. Determine the speed of the athlete.
- A. 10.0 m s^{-1}
 - B. 16.0 m s^{-1}
 - C. 25.0 m s^{-1}
 - D. 100.0 m s^{-1}

END OF PAPER

2

INDEX NUMBER

SIGNATURE

DATE: _____

**West African Senior School Certificate Examination
for School Candidates**

2 hours

[100 marks]

**For Examiner's
Use Only**

**Question
Number**

Mark

1. *In the spaces provided above, insert your **name**, full index number, normal signature and the date of examination.*
2. *This booklet consists of **sixteen** questions. Answer **all** questions in **Part A** and **four** questions from **Part B**.*
3. *Write the **question number** and your **index number** at the top of each page.*
4. *Write on **both** sides of the paper unless otherwise instructed on the question paper.*
5. *Begin **each** answer to a question on a fresh page. Leave two lines between answers where there are sub-questions to the same question.*
6. *On **no account** should you tear off **any** part of the booklet. It is an **examination malpractice** to do so. The entire booklet will be collected at the end of the examination.*
7. *Write in the space provided below, the **number of the questions** you have answered in the order in which you have written them.*

TOTAL

Answer all questions from Part A and four questions from Part B.

Credit will be given for clarity of expression and orderly presentation of material.

PART A

[40 marks]

All questions carry equal marks.

1. Explain how each of the following lifestyle factors negatively impacts female fertility:
 - (a) smoking;
 - (b) excessive alcohol consumption.[4 marks]
2. State four human practices that contribute to non-communicable diseases. [4 marks]
3. State the effects of each of the following activities on breathing:
 - (a) release of exhaust fumes from vehicles;
 - (b) deforestation.[4 marks]
4. Explain how each of the following practices could be used to prevent the spread of pathogenic diseases in a community:
 - (a) sex education;
 - (b) provision of potable water and good food.[4 marks]
5.
 - (a) Explain the fermentation process in the food and beverage industry.
 - (b) State two benefits of the fermentation process to humans.[4 marks]
6. Explain two ways in which each of the following gases is important to humans:
 - (a) carbon(IV) oxide;
 - (b) oxygen.[4 marks]
7. State two:
 - (a) benefits
 - (b) dangersassociated with the exploration of crude oil in Ghana. [4 marks]
8.
 - (a) State the conditions under which total internal reflection of light occur.
 - (b) Name two devices that operate on the principle of total internal reflection of light.[4 marks]
9. A vehicle of mass 2000 kg moving with a velocity of 90 m s^{-1} slows down to a velocity of 50 m s^{-1} in 20 s upon the driver seeing a pedestrian crossing the road ahead. Calculate the breaking force applied. [4 marks]
10. Explain briefly how a silicon chip may be made a better conductor of electric current. [4 marks]

PART B
[60 marks]

Answer four questions from this part.

- 11.** (a) (i) Explain the term *pathogens*.
(ii) Name two types of pathogens. [4 marks]
- (b) Consider the elements $_{13}\text{X}$, $_{6}\text{Y}$ and $_{14}\text{Z}$.
Classify **each** of the elements into their respective:
(i) groups;
(ii) periods. [6 marks]
- (c) Differentiate between a *real image* and a *virtual image*. [2 marks]
- (d) State **three** similarities between reproduction in plants and animals. [3 marks]
- 12.** (a) Explain four effects of disorders of the excretory system. [4 marks]
- (b) (i) What are *hydrocarbons*?
(ii) State two uses of the first member of hydrocarbons that exists in nature. [4 marks]
- (c) A convex lens of focal length 20 cm was used to produce an image of an object placed 50 cm from the lens.
(i) Calculate the image distance.
(ii) State two characteristics of the image formed. [5 marks]
- (d) Name two indigenous alcoholic beverages produced in Ghana. [2 marks]
- 13.** (a) Explain how the human developing embryo obtains nourishment. [5 marks]
- (b) A transformer has an input voltage of 400 kV. If the output voltage is 100 kV and the number of turns in the output is 5000, with a current output of 20 A, determine the:
(i) number of turns in the input;
(ii) the current in the input. [5 marks]
- (c) State two differences between the *physical properties* of acids and bases. [2 marks]
- (d) Describe briefly the preparation of dough for kenkey. [3 marks]
- 14.** (a) Explain the functions of **three** structures in the dermis of the human skin. [6 marks]
- (b) Define **each** of the following terms according to Bronsted-Lowry concept:
(i) an acid;
(ii) a base. [4 marks]
- (c) A learner intends to use a lens to burn a piece of paper held in their hands. At what point would the paper be held relative to the lens to produce the desired effect? [3 marks]
- (d) Name **one** element **each** that could be used to manufacture:
(i) electronic device;
(ii) electric cable. [2 marks]

15. (a) A ship with its luggage of cocoa beans weighs 55000 N and floats in sea water to a depth of h metres below the surface of the sea water of density 1030 kg m^{-3} .
- Name **two** forces that act on the ship as it moves through the water.
 - Determine the depth, h . [$g = 10 \text{ m s}^{-2}$] [5 marks]
- (b) (i) Name the component of air that is used in **each** of the following processes:
- Tissue respiration;
 - Photosynthesis.
- (ii) Give the systematic name of any **two** reagents used in the laboratory preparation of the component named in (i)(β). [4 marks]
- (c) State **three** features common to excretory organs. [3 marks]
- (d) Name **three** drugs commonly abused by adolescents in Ghana. [3 marks]
16. (a) Explain how *genetic testing* could be used to settle paternity disputes. [5 marks]
- (b) Consider an acid solution labelled 0.20 M HCl.
- Calculate the amount of solute HCl present in 25 cm^3 of the acid solution.
 - Determine the volume of 0.15 M NaOH that would be required to completely neutralize the HCl. [Na = 23, O = 16, H = 1, Cl = 35.5] [5 marks]
- (c) Describe **briefly** how electricity is generated from solar energy. [3 marks]
- (d) Give **two** reasons why exhalation is important to humans. [2 marks]

END OF PAPER

SC 5403
WASSCE 2026
GENERAL
SCIENCE 3
Test of Practical Work
1½ hours

3

CANDIDATE'S NAME	
INDEX NUMBER	SIGNATURE
DATE:	

THE WEST AFRICAN EXAMINATIONS COUNCIL

West African Senior School Certificate Examination
for School Candidates

SC 2026

GENERAL SCIENCE 3
TEST OF PRACTICAL WORK
[80 marks]

1½ hours

DIRECTIONS TO CANDIDATES

- Do not turn over this booklet until you are told to do so.
- In the spaces provided above, insert your **name**, **full Index number**, **normal signature** and the **date of examination**.
- Answer **all** the questions in this booklet.
- On **no** account should you tear **any** part of the booklet. The entire booklet will be collected at the end of the test.
- All rough work should be done in this booklet and then crossed out neatly. Under **no** circumstance should you work on **any** other paper.
- Write the **NUMBERS OF THE QUESTIONS YOU HAVE ANSWERED** in the order in which you have answered them, in the space provided below.

For Examiner's Use Only	
Question Number	Mark
TOTAL	

1. Fig. 1 (a) is an illustration of a system found in humans and Fig. 1 (b) shows the illustration of the structure of a functional unit of the part labelled IV.

Study the figure carefully and answer the questions that follow.

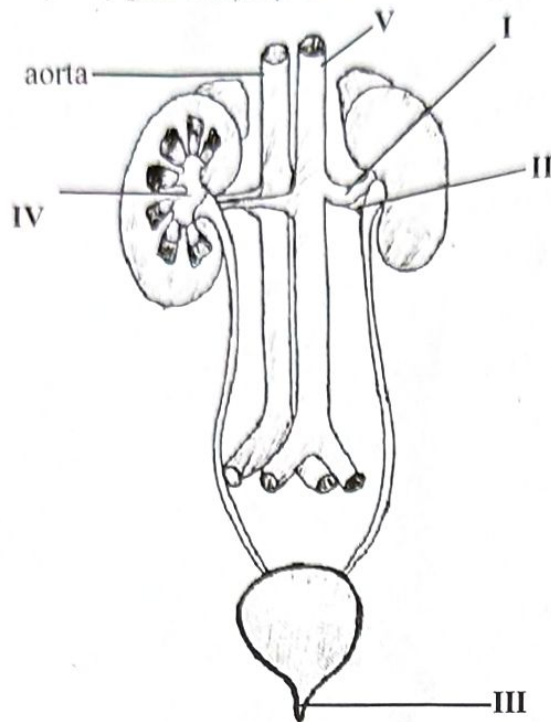


Fig. 1(a)

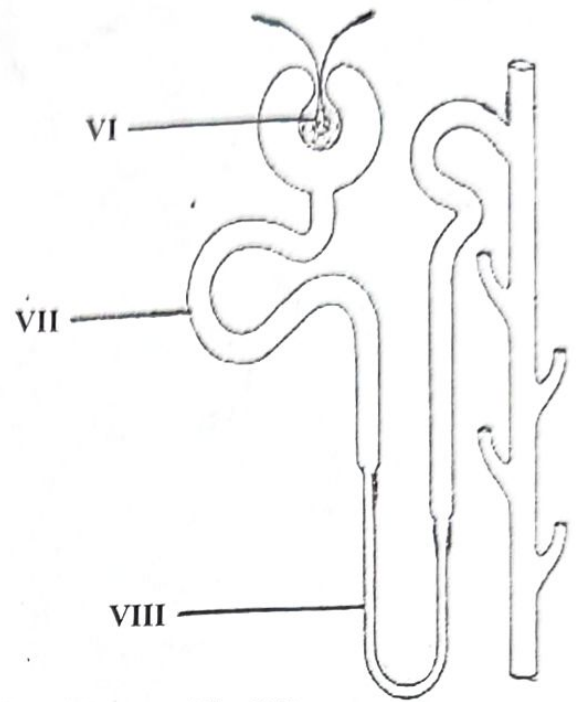


Fig. 1(b)

- (a) Name the:

(i) system illustrated in Fig. 1(a);

.....

(ii) functional unit illustrated in Fig. 1(b).

.....

[2 marks]

- (b) Name **each** of the parts labelled V, VI, VII and VIII.

V.....

VI.....

VII.....

VIII.....

[4 marks]

- (c) Explain the function of **each** of the parts labelled I, II, III, VI and VIII.

I.....

.....

II.....

.....

III.....

.....

VI.....

VIII.....

[10 marks]

(d) State **two** observable features of the part labelled:

(i) **VI**, which enhance its function;

(ii) **VIII**, which enhance its function.

[4 marks]

2. Fig. 2 is a model of pH-scale indicating the acidic and basic strengths of some selected chemicals used at home. The selected chemicals are potash solution, lemon juice, soap solution, table salt solution, hydrogen peroxide, hydrochloric acid and washing soda solution. These chemicals are represented as A, B, C, D, E, F and G, but **not** in any particular order.

Study the figure carefully and answer the questions that follow.

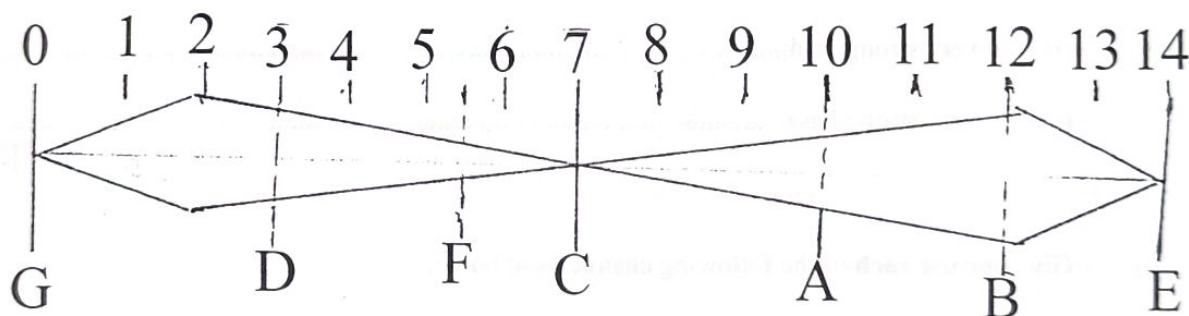


Fig. 2

- (a) Fill in the blank spaces in the table provided by assigning:
- each** of the named chemicals to the corresponding **correct** label;
 - the pH-value on the pH-scale;
 - the colour of the chemical upon addition of a universal indicator.

Label of Chemical	pH of chemical	Colour of Universal Indicator	Name of Chemical at home
A	10		Soap
B	12	Blue	
C	7		xxx
D			xxx
E	14	xxx	
F	xxx	Lime green	Hydrogen peroxide
G	0		

[8 marks]

- 4
- (b) (i) Select **two** chemicals listed which can react to give a product with $\text{pH} = 7$.
.....
..... [2 marks]
- (ii) Name the reaction in (i).
..... [1 mark]
- (c) Identify the chemical that **cannot** change the colour of red litmus and blue litmus.
..... [1 mark]
- (d) State the colour of methyl orange in:
- (i) **D**;
(ii) **E**..... [2 marks]
- (e) State the observation that would be made when **B** is added to:
- (i) **G**;.....
(ii) phenolphthalein [2 marks]
- (f) Identify the solution that could be classified as:
- (i) very strong acid;.....
(ii) very strong base..... [2 marks]
- (g) Give **one** use **each** of the following chemicals at home:
- (i) **B**;.....
(ii) **C**..... [2 marks]

3. In an experiment to investigate the effect of various solutions on potato cells, slices of potato were placed in equal volumes of **tapwater**, **distilled water**, and **salt water** for 48 hours. The slices were carefully removed from the solutions and their sizes measured again and compared.

(a) State **two** observations that would be made on the slices from **each** solution provided:

Tapwater;.....

.....

Distilled water;.....

.....

Salt water.....

.....

[6 marks]

(b) Explain the observations made in (a).

.....

.....

.....

.....

.....

.....

.....

.....

[8 marks]

(c) State the conclusion that could be drawn from the results obtained in the experiment.

.....

.....

.....

.....

[2 marks]

(d) Sketch and label a diagram to show the appearance of the cell placed in the:

(i) distilled water;

(ii) salt water.

[4 marks]

4. A learner wanted to use a piece of glass as a lens for the eyes. The learner was advised to determine the refractive index of the glass and was given optical pins and a protractor. The learner was directed to use any five angles of incidence and to find their corresponding angles of refraction. Fig. 4 shows angles of incidence $i_1^\circ, i_2^\circ, i_3^\circ, i_4^\circ$, and i_5° and their corresponding angles $r_1^\circ, r_2^\circ, r_3^\circ, r_4^\circ$, and r_5° . Study the Figure carefully and answer the questions that follow.

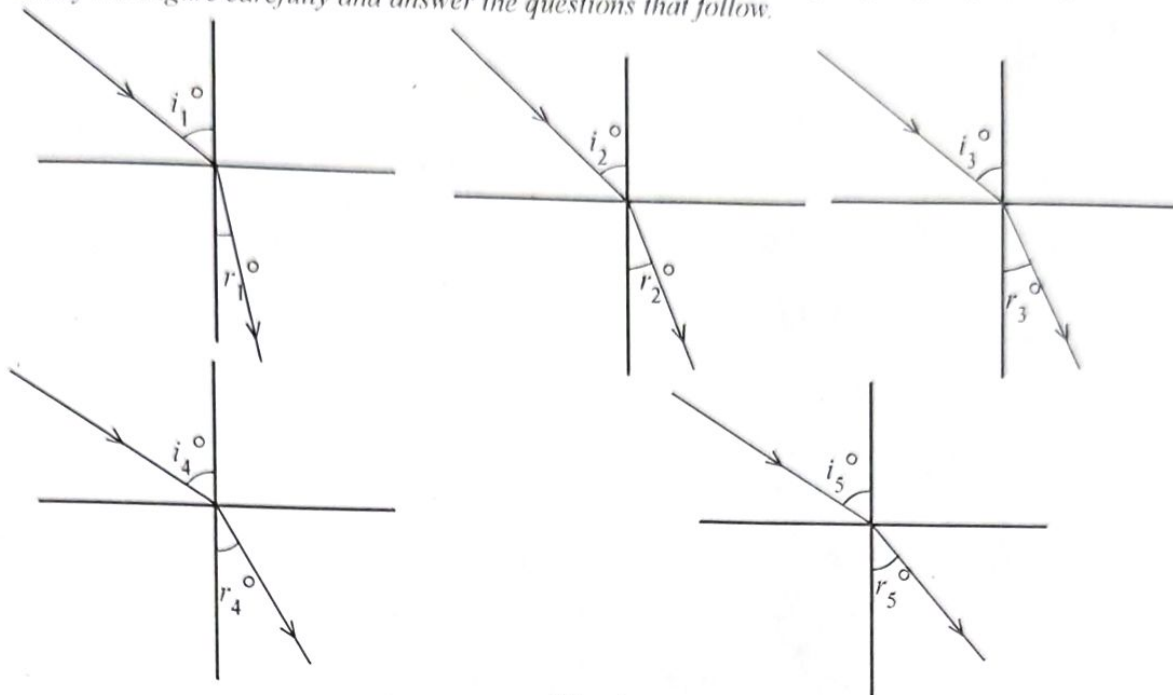


Fig. 4

- Measure and record the angles of incidence, $i^\circ = i_1^\circ, i_2^\circ, i_3^\circ, i_4^\circ$, and i_5° .
- Measure and record the corresponding angles of refraction, $r^\circ = r_1^\circ, r_2^\circ, r_3^\circ, r_4^\circ$, and r_5° .
- Determine the sine of the angles of incidence and their corresponding angles of refraction.
- Record the values obtained in the table provided.

Angle of incidence $i / ^\circ$	$i_1 =$	$i_2 =$	$i_3 =$	$i_4 =$	$i_5 =$
Angle of refraction $r / ^\circ$	$r_1 =$	$r_2 =$	$r_3 =$	$r_4 =$	$r_5 =$
Sin i	$\sin i_1 =$	$\sin i_2 =$	$\sin i_3 =$	$\sin i_4 =$	$\sin i_5 =$
Sin r	$\sin r_1 =$	$\sin r_2 =$	$\sin r_3 =$	$\sin r_4 =$	$\sin r_5 =$

- Plot a graph of $\sin i$ on the vertical axis and $\sin r$ on the horizontal axis.

[10 marks]

- Determine the slope of the graph.

[5 marks]

.....

.....

.....

.....

- State the significance of the slope.

[3 marks]

.....

.....

[2 marks]